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Learning processes during regime shifts: Empirical evidence from the diffusion of greywater recycling in Spain

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ABSTRACT

The water supply paradigm in urban Catalonia has experienced a significant transformation toward water decentralization since the approval in 2002 of the first ordinance on water conservation. Local water ordinances make mandatory the installation of rainwater harvesting and greywater reuse systems in new buildings. By 2012, about 50 municipalities in Catalonia had approved similar regulations. This paper analyzes the avenues and limits of the evolution and regionalization of the transition and focuses on the various types of learning that the adoption of greywater reuse systems has triggered in the Barcelona area. During the take-off of the technology, greywater reuse systems were relatively rudimentary but later on, membrane technologies gained popularity which facilitated a more widespread use and acceptance of the technology. The cooperation of the main actors involved in this water transition has been instrumental to move the transition forward but more efforts at the regional level are still required.

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1. Introduction: from centralized to decentralized supply and local water technologies

The use of large volumes of often distant, high quality water for the simple purpose of flushing the toilet and removing human waste is a reality in many households of the developed world. Today,

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alternatives to this old model of "flush and forget" are being sought to confront pressing environmental issues. As urban centers expand, concerns about water scarcity and water pollution arise. Options for reducing stress and uncertainty regarding urban water supply can be summarized under two broadly defined approaches. Supply can be enhanced by means of large-scale and centralized "technological fixes", be these dams, inter-basin water transfers, or more recent technologies such as desalination plants or wastewater and stormwater treatment plants. The large amounts of capital involved and the urgent need to enlarge water availability, combined with the dominant economic paradigms of early and mid-20th century (with Keynesianism having a hegemonic role in regulating economic life) led to the prevalence of centralized approaches to water supply. The urbanization process is key for understanding the origin and the evolution of this approach (Kallis and Cocossis, 2003), which has received different names, from the general command-and-control procedure and supply-side management to the more specific hydraulic paradigm (Del Moral Ituarte and Giansante, 2000), state hydraulic paradigm (Bakker, 2003), hydraulic age (Saurí and del Moral, 2001) or old hydraulic paradigm (Pedregal Mateos, 2005). Undoubtedly, water supply and sewage systems developed along those principles have produced large benefits to the population by improving the reliability of supply, reducing water-related diseases associated with poor water quality, and containing the impacts of extreme hydrologic events such as floods and droughts (Geels, 2006). On the other hand, they have also produced large costs, including ecological and environmental degradation, social disruption associated with infrastructure development, and economic and financial problems (World Commission on Dams, 2000; Gleick, 2003; Domènech et al., 2013a).

The eradication of water scarcity has traditionally been viewed as a techno-managerial problem, in which supply needs to be increased to meet rising demands; however, this approach may no longer be possible as a general solution (Guy and Marvin, 1996). Despite the fact that new infrastructures are and will be needed in some contexts, water managers are shifting their attention toward the other side of the equation, i.e. how society uses water (Gleick, 2003). This is known as water demand management, although a review of the existing literature leads us to conclude that there is not a common notion about what exactly water demand management is. Instead, we find a high range of conceptualizations. As Brooks (2006) contends, it is possible to unravel some simple definitions such as "getting the most from the water we have" or more elaborated ones, for instance defining water demand management as the "actions that reduce the amount of fresh water we use, or that keep water cleaner in the course of that use than it otherwise would be" (Brooks, 2006, p. 522).

The renewed interest in demand management strategies and more sustainable ways of dealing with water is in line with the growing appreciation for local water resources such as rainwater, greywater or underutilized groundwater. Rainwater harvesting and greywater reuse systems open a whole new window for urban water management, one in which end-users have a greater role to play. This alternative mode of water supply based on decentralization and on-site water recycling has gained relevance in recent years in some western countries such as Australia, Germany or Spain (Nolde, 1999; Brown et al., 2009). If decentralized water supply systems become widespread, desalination plants, dams and other large infrastructures will become less necessary and as a result, the current "urban water paradigm" may suffer significant transformations. At the same time, some actors such as maintenance operators and citizens will have to assume new roles and new values will have to arise (e.g. sustainability values will replace values related to security of supply) as well as new regulations and water quality standards will need to be created.

The notion of socio-technical, sustainability transitions (van den Bergh et al., 2011) serves as a useful framework to describe the system-wide changes and learning that need to take place for these new technologies to find their place in the dominant hydraulic water regime. Such transformations to build more sustainable societies need to take place at different levels and scales. In that sense, and building on the concepts of niche, regime and landscape, socio-technical transitions can be approached from a multi-level perspective where dynamic changes at different levels jointly take place (Rip and Kemp, 1998; Geels, 2002; Geels and Kemp, 2007; Smith and Stirling, 2008). Technical and social elements are linked in such a way that they provide stability to a particular socio-technical regime (e.g. the centralized water model of "flush and forget"). In this paper we adopt Geels (2002) definition of a socio-technical regime as the set of rules that structure and guide the activities of social groups. In this context, radical innovations take place at the micro-level, in niches, where new developments receive

certain protection against the regime (Geels and Kemp, 2007). Niches provide a space for learning processes, as initially new technologies may experience frequent failures and they may be unable to meet users' needs satisfactorily. Niches and regimes are embedded in a socio-technical landscape. The landscape metaphor refers to a series of external factors such as economic growth, cultural values, political coalitions, environmental problems or resource scarcity that influence the development of a socio-technical transition process (Geels and Kemp, 2007). Landscape changes create pressures and opportunities for niches to develop, and at the same time, they may set back transformation processes.

To abandon socio-technical regimes based on unsustainable technologies – such as centralized large-scale systems- is not easy (Dangerman and Schellnhuber, 2013) because the actors in control of the regime are powerful and deep-rooted social norms need to change. Along these lines, van den Bergh et al. (2011, p. 14) argue that "much transitions research assumes that transitions are rather smooth, well-governed processes. But like all social and technological changes, transitions involve creative destruction with winners and losers." The role of civil society and public authorities becomes essential to promote sustainability transitions since private actors alone (e.g. water supply companies) may have little incentives to achieve the "collective good" of sustainability (Geels, 2011). At the same time, the benefits of less environmentally damaging technologies are not always obvious to the user. Small and decentralized technological systems may be less efficient in economic terms and need an expertise still limited in comparison to large systems (Domènech et al., 2013b). All in all, citizens may need to renounce to part of their constructed notion of comfort and assume greater economic costs (and invest personal time as well) for the common goal of sustainability (Chappells and Shove, 2005).

The objective of this paper is to study the learning dynamics and scaling-up opportunities behind the ongoing emergence of a new urban water regime based on the use of decentralized water technologies. We argue that learning processes determine the direction and speed of the transition. More specifically this paper focuses on the promotion and adoption of greywater reuse systems at the niche level, in the Barcelona area (Spain). Greywater reuse systems present significant distinctive features to the conventional centralized water supply systems in which a central authority distributes nearly pristine water to the users. People tend to turn on the tap, either drink or pollute most of the water consumed and then, forget about the consequences. This behavior needs to change with the use of greywater systems. And for this reason, our analysis mainly focuses on these systems, although most findings are also applicable to the case of rainwater harvesting.

Greywater systems enable the reuse of low-polluted water on-site through a fairly simple treatment process. In the Barcelona area, greywater reuse systems typically involve reusing water from the shower or the bath for toilet flushing purposes. Greywater reuse has certain features of interest for urban water managers: greywater produced in the shower or the bath has a higher quality than wastewater from other water uses in the household; treatments are generally not costly, and, when compared, for instance, with rainwater harvesting, the quantity and reliability of this source is fairly high. In sum, we could think of greywater as high quality wastewater with many potential applications in the household (Eriksson et al., 2002).

The adoption of a multi-level perspective (MLP) (Geels, 2002) is useful to emphasize the critical importance of the existing regime and socio-technical landscape in shaping the success or failure of greywater reuse, in addition to the processes taking place at the niche scale. The transition under study can be described as "goal-oriented" or "purposive" (Smith et al., 2005; Geels, 2011) since environmental and technological-specific policies have been instrumental for the transition to happen. However, we show that traditional practices of command and control alone (i.e. regulations) are not sufficient to guarantee a successful change of regime. Transition management practices are also necessary to influence and steer the direction of the transition. Key elements of transition management, among others, include learning-by-doing through experimentation, doing-by-learning, and participation, communication and interaction between stakeholders (Fiorino, 2001; Kemp and Loorbach, 2006). In this paper, we focus on the co-evolutionary processes of learning-by-doing and doing-by-learning which regard the development of theoretical knowledge from practice and applying this knowledge again to future actions through long-term strategic visions and goals (Kemp et al., 2007).

Learning is a broad concept that is subject to different interpretations depending on the context and the purpose of action. In general terms, learning can be defined as the process of assimilating new information. Different types of learning have been identified in different contexts such as policy

learning, social learning, institutional learning, and organizational learning. Learning frequently emerges as a result of past experience which can subsequently be applied to future actions and thus, becomes a driver of change and improvement (van der Kerkhof and Wieczorek, 2005).

Much learning is generated throughout the different stages of a transition as a result of a reflexive learning cycle. Individual learning is not sufficient to move toward sustainable development. In that sense, collective learning processes that result from the interaction and communication between stakeholders are critical to understand societal adaptation and innovations (Röling and Wagemakers, 1998). Some authors differentiate between single, double and triple-loop learning. Single-loop learning (or instrumental learning) relates to the cognitive level of analysis and concerns "new insights into the policy options in the case of a given policy problem and a given policy context" (van der Kerkhof and Wieczorek, 2005, p. 735), or in other words, the consequences of specific actions (Reed et al., 2010). Double-loop learning (or transformative learning) relates to the normative level of analysis and concerns "new insights not only into the solutions to a certain problem but also into the problem itself and the context in which decisions take place" (van der Kerkhof and Wieczorek, 2005, p. 735). Eventually, third-loop learning involves "changing the values, norms, and higher order thinking processes that underpin assumptions and actions" (Reed et al., 2010)

For the purposes of this paper, however, we will draw on the framework developed by Fiorino (2001) regarding three main types of policy learning, differentiating between technical, conceptual and social learning (Fiorino, 2001; Farrelly and Brown, 2011). In the first place for technical learning (or first-order learning) we understand the "outcomes associated with implementing and refining technologies and policy instruments"; in the second place, conceptual learning (or second-order learning) "involves questioning (reconceptualizing) the fundamental policy and objectives (the status quo)", and thirdly, social learning (or third order learning) "provides the opportunity and levers to promote regular shifts (transformation) in the sophistication of learning, from technical to conceptual" (Farrelly and Brown, 2011, p. 723). Concerning the latter, Reed et al. (2010) define three main conditions that social learning processes should fulfill: (i) they should involve a change in understanding at the individual level; (ii) should go beyond the individual and become situated within wider social units or communities of practice and (iii) occur as a result of social interactions and processes between actors within a social network.

The emergence and establishment of greywater technologies in the Barcelona area provides a good example of these three levels of learning, all critical to push the transition forward. In what follows, we first briefly sketch out the recent changes in water politics at the Spanish level as well as the recurrent drought episodes in Catalonia, which have put against the ropes the traditional water supply system. In Section 3, we make a brief summary of the evolution of local and regional regulations that promote, among others, the adoption of greywater reuse technologies in the Barcelona area. In Section 4, we present the methodology. In Section 5, we first assess such a transition and the parallel learning process, with special emphasis on the municipality of Sant Cugat del Vallès, a wealthy suburb of the Metropolitan Area of Barcelona and the niche where alternative technologies were first implemented and much technical learning was generated. Then we analyze the hurdles and promising steps toward the generalization of the use of this "new" resource; this allows us to reflect on the complex entanglements of actors underpinning such transitions. We also show that while technology may have a basic role to play in order to enable a sustainable transition in water resources management, it is in the sphere of social learning where efforts should be made.

2. Contextualizing the emergence of local water alternatives in Barcelona: changing national water politics during the 2000s

The emergence of local water initiatives challenging the hegemonic supply-led model, deeply entrenched in Spain and Catalonia, has been the result of the interaction of multiple actors and factors at different scales. Using the terminology of transition studies we could argue that "landscape developments" taking place at different scales have exerted an important influence on the trajectory of alternative water resources in Metropolitan Barcelona. Among many, we argue that changing water politics and policies at the Spanish and Catalan levels, as a result of political changes and also the

increase in drought frequency, favored the proliferation of alternative water supply systems in the Barcelona area.

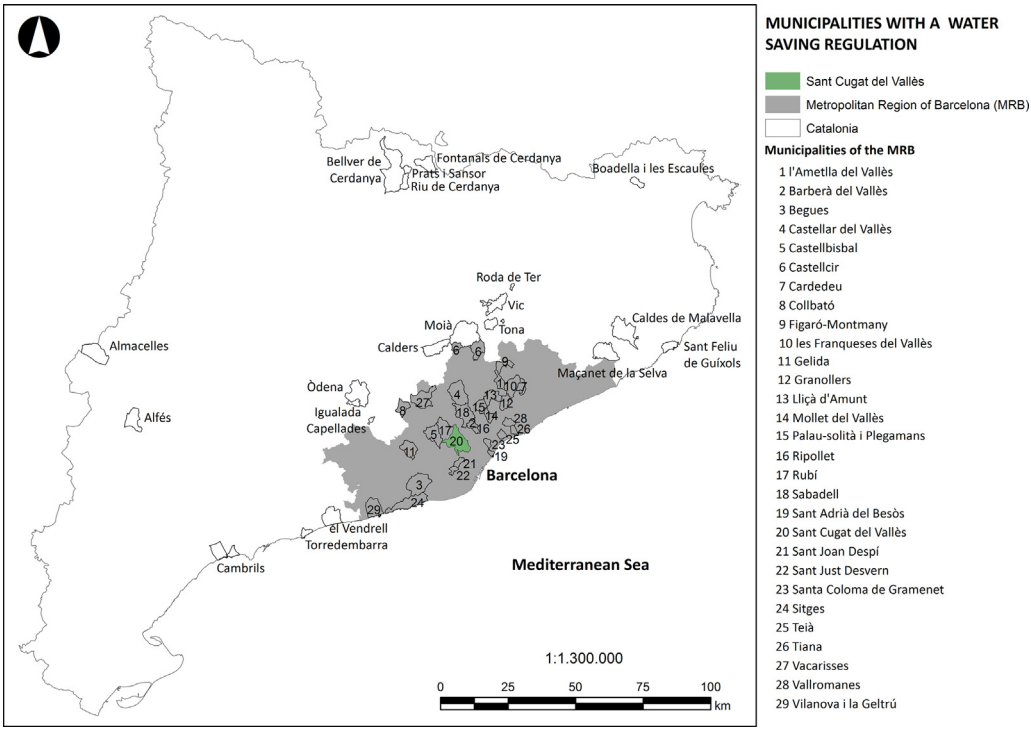
Spanish water politics in the 20th century may be read as a paradigmatic example of the hydraulic paradigm mentioned beforehand. Massive centralized works, including dams and water transfers, have been the main axis of the dominant socio-technical regime of water management during the past century. The National Water Plan (PHN, *Plan Hidrológico Nacional*) approved in 2001 was a clear expression of such paradigm, including an interbasin transfer of some 1050 Gl from the Ebro River to Eastern and Southern Spain and to the Barcelona area as well (see Masjuan et al., 2008). The PHN, however, generated much social opposition, and the mobilization of civil society was widespread, with big demonstrations in the streets of Barcelona, Zaragoza or Madrid, and a "Blue March" toward Brussels. Most of the academic and scientific community also opposed the project and this opposition coalesced in what was later called *Nueva Cultura del Agua* (New Water Culture) around which converged the broader social movements mentioned above. The socialist victory in the Spanish elections of March 2004 implied an important shift in water resources planning and policy in the country, which were translated into the canceling of the Ebro transfer. The socialist government adopted partially the postulates of the New Water Culture, and emphasized demand management measures. Nevertheless, a supply-side alternative was also projected in order to avoid conflict in the zones that would have been watered by the Ebro transfer; desalination was the option chosen, and several plants were planned and developed along the Mediterranean coast, including one in the Barcelona metropolitan area.

In the first decade of the 21st century, water problems have been aggravated. The years 1998–2002, 2004–2005, and 2007–2008 have observed water supply falling to critical levels to the point that eight drought warnings were issued by Catalan authorities in this period. The last episode of drought (2007–2008) was the outcome of the worst precipitation deficit recorded since the mid-1940s (Pastor, 2008). As March et al. (2013) documented, some twenty months with little or no rain led reservoirs supplying Barcelona to fall to one fifth of their capacity. The extreme gravity of the situation obliged the Catalan Water Agency to ban some uses such as public fountains and public parks watering, and also the use of water for private gardens and swimming pools. It also forced the shipments of water to Barcelona by sea tankers or the building of pipes to distant rivers. Beyond these actions, awareness campaigns aiming to voluntary reductions in consumption were also launched. In terms of water demand and according to metropolitan and water authorities, the different droughts episodes have contributed to a decline in water consumption per capita in Metropolitan Barcelona (ACA, 2010). The aftermath of the drought, as March et al. (2013) argue, not only left 490 million Euros in economic costs of emergency measures but also brought to the fore the ongoing debate between the new water supply solutions and the old "hydraulic paradigm" of inter-basin transfers.

In spite of not completely breaking with the entrenched hydraulic paradigm, this shift in water politics and the New Water Culture movement, directly or indirectly, contributed to a change in how local resources were imagined. The growing interest of local authorities for environmental issues was also strongly stimulated by the development of Local Agenda 21 Plans in many Spanish and Catalan municipalities during these years.

3. The rising interest for local water resources: the role of local ordinances in boosting the emergence of a new socio-technical regime

In 2002, the municipality of Sant Cugat del Vallès, in metropolitan Barcelona, became a pioneer in local water resources management after approving the first water saving ordinance of Catalonia (Sant Cugat del Vallès, 2002, 2008). A salient feature of the ordinance was the requirement for all new buildings with a minimum of eight apartments to install a greywater reuse system. Water from the shower would be collected and piped to the basement of the building, where a special tank would perform a basic on-site treatment and disinfection using chlorine. Once treated the water would be pumped back to fill the toilet tanks of the different apartments. The regulation also included the requirement for new buildings with a given garden area to install a rainwater harvesting system.



Map 1. Municipalities with a water saving ordinance in Catalonia.

Soon after, many other municipalities approved similar ordinances, and by 2005, fourteen municipalities in Catalonia had local regulations concerning water. All the ordinances had a similar spirit but requirements and measures differed to suit different local conditions. In an effort to assist other interested municipalities to develop their own ordinance, in October 2005, the *Diputació de Barcelona* (Provincial government) approved the "Framework Ordinance for Water Savings" that would serve as a guide for the development of new local ordinances. In 2006, national (Spanish) and regional (Catalan) legal instruments enforcing the installation of water saving devices in new buildings were also approved. In the Spanish Building Code (*Código Técnico de la Edificación*) there was no reference to the installation of greywater reuse systems while in the Catalan Eco-efficiency Decree (*Decret d'Ecoeficiència*) greywater reuse was only listed as an eligible measure. In addition, the Catalan River Basin District Management Plan (*Pla de gestió del Districte de conca fluvial de Catalunya*) approved in 2010 stated that the Health Department, the Environmental and Housing Department and the Catalan Water Authority would work together to establish criteria to regulate the use of greywater at the regional level.

At the end of 2011, more than 50 municipalities in Catalonia (totaling 1.2 million people) had adopted a water saving ordinance (Map 1). The droughts experienced by the Barcelona area in 2005 and 2007–2008 increased the pace in the approval of water saving ordinances as shown by the peak in the publication of ordinances after these episodes (Fig. 1). Perhaps more importantly, the adoption of water saving ordinances appeared systematically in the recommendations arising from Local Agenda 21 processes involving many municipalities during these years.

In any case, these pioneering regulations which are transforming the configuration of the urban water cycle constitute a good example of the emergence of sustainable practices at the local level. The equivalent regional and national regulations (Spanish building code and Eco-efficiency Decree mentioned before) are by far less ambitious, which suggests that promoting sustainability practices through a top-down approach may be more difficult to pursue.

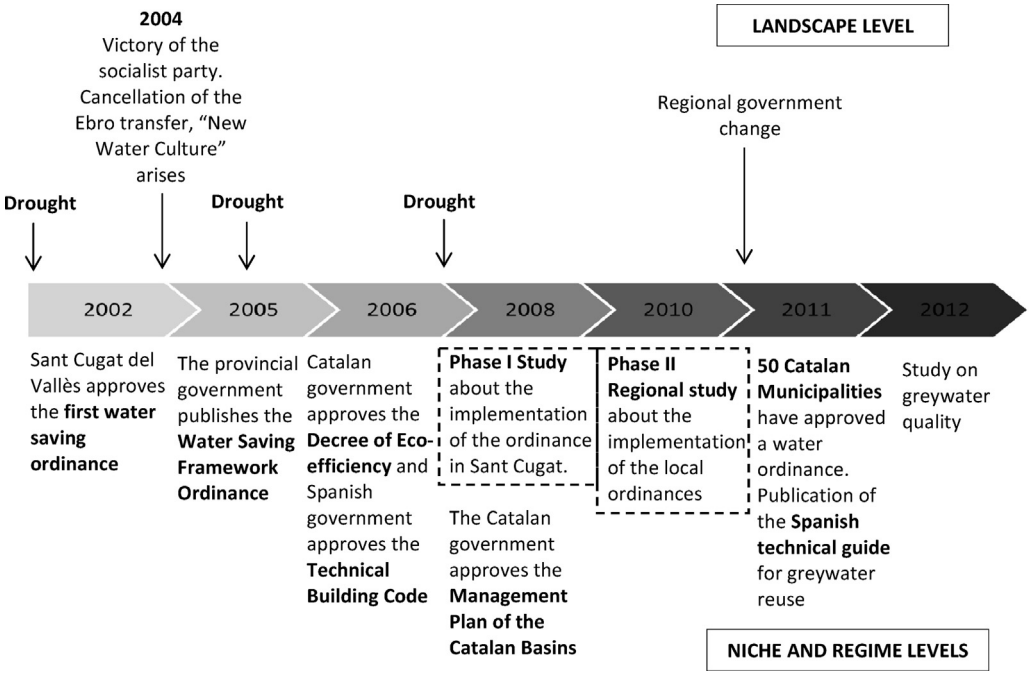


Fig. 1. Time-line of the main events of the water transition at the different levels.

4. Methodology

The emerging urban water regime involving greywater reuse has been analyzed with a mixture of methodological tools based on a case study research. The study consisted of two different phases. In phase I, we conducted our research in the town of Sant Cugat del Vallès to study the transition processes and learning generated at the niche level. As said before, this municipality was the first to approve in 2002 a water saving ordinance that, among other things promoted the use of greywater reuse systems. In 2008 we conducted a survey among 120 greywater users in this municipality in order to analyze users' perceptions and acceptance of the technology after the first five years of implementation of the ordinance. We also interviewed rainwater harvesting users, installation and maintenance companies, building administrators and developers (Table 1). The results of the first phase of the study are used in this paper to explain the learning generated and the main impediments encountered at the niche level. In late 2008 and early 2009, three seminars led by the Catalan Water Authority were held with the main actors involved in the promotion of these new technologies in order to discuss the main findings of the study.

Table 1
Stakeholders interviewed or surveyed during the two-phase study.

Stakeholders	Number	Phase of the study
Greywater users in multi-family buildings	120	1
Rainwater harvesting users in multi-family buildings	20	1
Rainwater harvesting users in single-family buildings	13	1
Building administrators	2	1
Installers and maintenance contractors	10	1
Developers	2	1
Building administrators	2	1
Environmental and urban planning managers of the municipality	24	2

In order to identify the learning factors that may favor or hinder the break-through of the new water technologies to the regime level, we conducted in 2010 the second phase of the study in which all the Catalan municipalities with a water saving ordinance were included. In this second phase, we compared systematically the ordinances approved by each of the aforementioned fifty municipalities. We also conducted fourteen key-informant interviews with the environmental and urban planning managers of the municipalities that had more experience with the implementation of the regulation, and among the rest of municipalities, we conducted an on-line survey that was answered by ten municipalities. In total, taking into account the interviews and the survey, we obtained consistent information of twenty four municipalities (48% of the sample). The study concluded with a seminar supported by the Provincial Government for local environmental and urban planning managers to discuss the results of the interviews and surveys. In total, thirteen municipalities participated in the seminar (10 with ordinance and 3 without ordinance).

Data from the interviews were analyzed through qualitative content analysis to identify the main themes and types of learning that relate to each actor and transition stage. Data from the first phase of the study have been used to analyze the technical learning generated during the first years of implementation of the regulation. The second phase of the study has provided evidence of the second-order of learning or conceptual learning. Finally, both studies have contributed to social learning processes with a large number of actors at local and regional scales involved.

5. Results and discussion

The transition started by Sant Cugat del Vallès in 2002 has proven rather successful since in a period of less than ten years another fifty municipalities in Catalonia (with a total population of about 1.2 million people) have approved similar regulations. Water saving ordinances involve many possible actions but the development of decentralized and alternative technologies has become a key aspect of the ordinances. Rainwater harvesting and greywater reuse became mandatory in most new buildings of 48 municipalities and 32 municipalities respectively. The smaller number of ordinances promoting greywater reuse in comparison with rainwater harvesting lies in the fact that greywater reuse was not considered in municipalities having a predominantly disperse urban form. This is because greywater systems are less cost-effective in single-family houses than in apartment blocks due to the smaller volume of water reused. Besides that, some local environmental managers and other actors are still reluctant to enforce greywater reuse because they have concerns regarding the potential and performance of this technology. For instance, one of the municipal urban planning managers interviewed (Phase II) expressed his concerns in relation to the installation of greywater reuse systems in his municipality: “The municipality decided not to include greywater reuse systems in the regulation due to the health risks involved when the maintenance of the system is not satisfactory. The municipality does not want to assume such responsibility” (Interview #8, 21 October 2010).

Several obstacles still have to be overcome and much learning needs to be generated if these technologies want to become part of the urban water regime. In particular, fewer municipalities have included greywater reuse technologies in the water saving ordinance which reflects the various challenges and demands posed by this particular technology. In the next subsections, we discuss the various types of learning that the adoption of greywater systems has triggered in the Barcelona area, starting from the niche level (Sant Cugat del Vallès).

5.1. Technical learning at the niche level

In what follows, we analyze the main technological developments taking place in the town of Sant Cugat del Valles as a result of the first years of experience with greywater technologies and we link this knowledge development with user’s reactions and attitudes. The first greywater systems in this municipality were installed in apartments for young people subsidized by the municipality. The systems were quite simple and usually involved a basic filtration device to capture coarse elements and a tank for the chlorine treatment. Maintenance required the removal of refuse from the filtration devices and the periodic replacement of the chlorine pill (usually once a week). Later on, after the approval of the water saving regulation, several buildings adopted the same type of technology. While

some developers selected slightly different greywater treatment technologies. Greywater systems raised mixed feelings among flat owners or tenants. According to the results of the survey conducted among 120 greywater users in 2008, more than 50% of the users declared themselves to be satisfied or very satisfied with having a greywater system. However, some 35% were unsatisfied or very unsatisfied with the system (Domènech and Sauri, 2010). A rather usual scenario was one of satisfaction at the beginning (in part related to the more pro-environmental attitudes of dwellers) that turned into dissatisfaction when the first problems in maintenance appeared. The use of chlorine, in particular, proved very problematic because of lack of knowledge in proper doses and the need for a continuous maintenance. If chlorine pills were not adequately replaced, foul odors would rapidly expand and the appearance of water deteriorated quickly as well. Likewise, excess zeal in adding chlorine also increased sensations of discomfort. While the first maintenance services sometimes performed a deficient job, the behavior of contractors and builders did not help much either. During the first years of operation of these systems the environmental department of Sant Cugat city council had to argue and sometimes fight hard with contractors in order to have the systems properly installed and maintained.

However, the technology evolved as a result of the knowledge generated during the pre-development stage of the transition. In Sant Cugat del Vallès and ten years after the approval of the local ordinance, more than 5000 households have a greywater reuse system. Recent greywater systems appear to have overcome many of the problems of the first installations. As in other parts of the water cycle, a major breakthrough has been the proliferation of systems using extremely fine filtration procedures in the form of membranes. The most advanced systems installed in the municipality in 2010 and 2011 are based on a much more sophisticated and reliable technology. For instance, in one of the systems currently used in new housing developments water coming from showers is subject to a double process. First, a biological cleansing using bacterium allows for the destruction of all organic matter. Second, the resulting water is subject to a process of ultra-filtration with a pore size of 0.05 μm. In this way, no further chemical treatments are needed. Instead of the weekly or more frequent maintenance tasks, these newer systems can operate for months, and maintenance is performed by a specialized company. Hence, many of the problems associated with the early greywater systems pointed above have disappeared, especially those related to odors and the management of chemicals.

We observe that the accumulated experience (technical learning) along the years has improved the functioning and management of systems and has also informed other processes of learning (conceptual and social).

5.2. Conceptual learning: the role of the municipality in reinforcing the transition

Given that the transition is goal-oriented and is mainly led by the municipalities through the local ordinances, we have analyzed in detail, the conceptual learning generated by the municipalities at the policy level as well as the main impediments that the municipalities have to overcome to favor a successful transition toward water decentralization.

The systematic analysis of the text of the 50 regulations approved until the end of 2011 reveals that the content of the regulations and the technical guidelines for the design of greywater reuse systems varies significantly across municipalities. This lack of regional standards regarding greywater reuse creates insecurity among installers and users and, at times, may even hinder the adoption of this technology. For example, most ordinances state that greywater reuse systems should be designed according to the volume of water produced (from showers/baths) but the accumulated experience (technical learning) has proved that it is more optimal to adjust the system to the planned water demand (from toilets/gardens) in order to avoid the accumulation of stagnant water in the deposit. So far, this latter option is only found in three ordinances (one of the last ordinances published in 2010 and two modifications). The fact that municipalities are changing this parameter in their regulations is a positive indicator of the learning process. The Spanish technical guide for greywater reuse (published at the end of 2011) also recommends considering the final demand when designing the system (AQUAESPAÑA, 2011). However, there is also lack of consensus on the reference value for calculating the water demand from the toilet, which ranges from 18 to 48 l/person day depending on the ordinance. As one of the environmental managers interviewed pointed out: “The text of the water saving

regulation should be simple and comprehensive, so everybody can understand it. The technical stuff should be included only in annexes (. . .) there should be a water saving regulation at the regional or state level" (Interview #4, 2 September 2010). In total, at least ten municipalities have modified the initial regulation approved to incorporate the experience acquired during the transition process. However, the approval of a water saving regulation at the regional or state level still seems somehow far away.

Problems with the implementation of greywater systems and the ordinance in general appeared also internally in the municipalities as stated by various local managers of the municipalities interviewed. One first constraint faced during the elaboration and implementation of the ordinances was the lack of coordination and cooperation between the different departments. The ordinances are usually developed by the local environmental department, but frequently, it is the urban planning department which has oversight of the implementation of the ordinance. Unless the urban planning department understands and agrees with the benefits and requirements of the regulation, the regulation cannot be implemented effectively. This is a classic example of the mismatch between urban and environmental policies (see, for example, Bai et al., 2010). As Fiorino (2001) points out an important component of conceptual learning is the development of integrated strategies.

Another important step of the implementation process concerns communication with the affected public. Citizens become direct users of these systems and therefore, need to be aware of system requirements (e.g. chemical products thrown down the bathtub can harm the system). Users also need to know that recycled water is being used for the toilet or the garden and in view of that, take the required precautions. Municipalities have adopted various strategies to inform citizens about the new regulations and the specificities of decentralized water systems. Environmental and urban planning managers of the municipalities with a water saving ordinance were asked about the communication strategies used to disseminate the content of the ordinance: 88% of the municipalities published the ordinance in their website or local bulletin, 67% inserted news in the local media, 46% distributed water saving devices, 33% conducted informative sessions, 25% produced handbooks about the ordinance, and 17% set up an information booth. However, the results of the survey conducted with the users of the greywater systems point at a limited efficacy of these actions. Survey results revealed that over 50% of the users ignored basic information about the operation of the system while also over 50% of users declared not having received enough information about greywater reuse (Domènech and Sauri, 2010). Media campaigns at the regional level, quite successful in stimulating conservation actions during the drought periods, could be an effective way of spreading the message and reaching a greater proportion of the population.

Another important concern for the municipality is how to monitor the operation of decentralized systems. In general, ordinances state that developers need to sign a maintenance contract with a professional company with two years of duration minimum. Likewise, facilities would need to be monitored to detect possible malfunctioning of the systems. The municipality supervises and authorizes the startup of the systems when these are new. After that, in general, city councils do not supervise the operation of the systems because they lack the capacity and human resources required to do so on a regular basis. In that sense, one of the environmental managers interviewed wondered: "Up to what point can we control if the systems are operating correctly? Is this municipality's responsibility? If so, we should train the municipality staff, so they can undertake this task" (Interview #10, 25 October 2010). Another manager from another municipality added: "It is very complicated because you are invading the civil sector and a private property. We can obligate them to make a maintenance plan, but we cannot check whether it is executed or not" (Interview #8, 21 October 2010).

Traditionally, water supply companies (either public or private) have been responsible for supplying drinking water through centralized networks. However, in this new scenario of water supply decentralization citizens have a greater role to play. Users' actions have an immediate impact on the greywater system and the resulting water quality. Users are also the first to notice and suffer any disruption of the system and therefore, they may need to take action and contact the company responsible for maintenance directly. Other strategies to improve the monitoring of the systems are still under consideration by the municipalities. In order to improve the control over the operation of greywater systems, the environmental manager of one of the municipalities interviewed sees a lot of potential in remote control systems (Interview #1, 27 August 2010). Some companies are indeed starting to

Table 2
Main difficulties identified and proposals suggested by the environmental managers of the municipalities.

Technical difficulties	**Proposals**
Sizing greywater reuse systems (tanks)	Flexible formulas for calculation Calculate tank size according to water demand
Lack of maturity of technology and lack of professional experience	
Small business market	
Submitting the right documentation to get the housing permit and to apply the ordinance	Provide models of a maintenance contract, inspection sheet, inventory of measures, etc.
Economic difficulties	**Proposals**
Lack of inspection capacity of city councils	Create a regional certification authority
Lack of visualization of the economic and environmental benefits of the water saving ordinance	Install meters to record drinking water savings Design a water bill to encourage savings
Uncertainty regarding the economic and environmental viability of decentralized systems	Conduct life cycle analysis, to analyze the most effective tank size, to modify the price of water, etc.
Other difficulties	**Proposals**
Lack of users' awareness about the content of the ordinance and lack of information	Carry out awareness campaigns about the ordinance to citizens and also to the stakeholders involved in its implementation
Doubts about how to interpret the ordinance	Increase the coordination of the different municipal departments involved in the implementation of the ordinance Create a technical advisory committee
Overlapping of regulations (ordinances, Spanish Building Code and Decree of Eco-efficiency) and competences (local, regional and state)	Unify criteria

explore this alternative. Another alternative proposed by an environmental manager is the creation of a regional entity to supervise the correct functioning of the systems following the successful model already in place for elevators (Interview #2, 30 August 2010).

As a result of the conceptual learning developed in the Barcelona area during the first years of implementation of the regulations, the problems that greywater reuse systems are facing are better understood and some strategies to solve them are starting to be debated (Table 2). In order to scale-up the use of decentralized water systems and overcome some of the constraints exposed, the collaboration of the main stakeholders involved in this process is critical. In the next subsection we focus on progress made in this area.

5.3. Steps toward reinforcing the transition: social learning and the role of regional institutions

The agency of regional and sub-regional institutions has been critical to reinforce the transition process and make the first steps toward its regionalization. The provincial government together with the working group "new water culture" of the "Network of Cities and Towns toward Sustainability" formed by nine Catalan municipalities developed a framework water saving ordinance in 2005 which was a starting point to reach some consensus about the main features of local regulations. The framework ordinance has been a valuable reference for many municipalities, although currently, the ordinance could benefit from a revision process. In 2008, the Catalan Water Authority financed a research study to evaluate the implementation of the local water ordinance in Sant Cugat del Vallès with an important focus on greywater reuse (Domènech et al., 2009). As explained before, this paper draws on some of the data and results of that study (Phase I). After the completion of the study, a series of meetings with the main actors of the sector were held to discuss the results of such study. These meetings were critical to debate the need to introduce changes in the delivery process and the regulations themselves, and to advance the conceptual learning processes for successfully scaling-up the new technologies at the regime level. The involvement of the Catalan Water Authority facilitated the mobilization and close collaboration of the main actors of the sector. After these meetings, the main companies responsible for developing greywater reuse systems – all belonging to a professional association called AQUAESPAÑA

– agreed to create a working group with the objective of elaborating a series of technical recommendations about the design and operation of greywater systems. In 2011, the first Spanish Technical Guide for greywater reuse was published, and a technical session was organized to present the guide and discuss some of the contested issues related with greywater reuse. The meeting included presentations from the Catalan Water Authority, the *Diputació de Barcelona*, the municipality of Sant Cugat del Vallès, companies, installers and researchers.

The provincial government (*Diputació de Barcelona*), and particularly the so-called "Network of Cities and Towns toward Sustainability" in close contact with other local environmental networks in Europe was another important actor able to influence the course of the transition. Besides elaborating the framework ordinance in 2005, the provincial government organized several meetings with the environmental managers of the municipalities and in 2010 funded a research study to analyze the implementation process of the local regulations (Vallès et al., 2011). Again, this paper draws on some of the data and results derived from that study (phase II). At the end of the study, the provincial government organized a seminar to discuss the results of the study with the environmental managers of the municipalities which again was a good means to stimulate conceptual and social learning. The provincial government was planning to update the framework ordinance and incorporate some of the lessons learnt from the study and the various meetings held, but to our knowledge, this has not been materialized yet.

Collaboration between different entities in an attempt to reconceptualize existing policies related to greywater reuse also involved the Regional Health Department, the Catalan Water Agency and the working group of AQUAESPAÑA. The objective in this case was to define water quality standards for greywater reuse. As a result of this collaboration, these institutions conducted a study with the support of Sant Cugat del Vallès municipality to measure greywater quality considering different treatment processes and different stages of the recycling process. The results of the study provided useful technical learning to be used in future policy development. Some systems were not working at the optimum level since turbidity of treated greywater was high. *Escherichia coli* was effectively removed in all systems but *Clostridium perfringes* showed resistance to some treatments (Generalitat de Catalunya, 2012). In spite of the efforts of these three organizations and the fact that water quality control is an extremely sensitive issue in greywater reuse systems, to our knowledge no water quality standards for greywater reuse have been approved yet by any competent authority in Catalonia or Spain which suggests that social learning processes still need to take place at this level. In the guide published by AQUAESPAÑA there is a section with recommendations about water quality standards and frequency of analyses (AQUAESPAÑA, 2011). However, it is unlikely that these recommendations are effectively adopted since the cost of the analysis remains high and therefore, greywater reuse technologies would become less viable if these tests became mandatory. Given the impasse at higher levels, some municipalities are planning to publish an annex in their own water ordinance with water quality standards.

According to Fiorino (2001, p. 324) social learning "builds on the cognitive capacities of technical learning and the rethinking of objectives and strategies that occurs in conceptual learning" but it also "emphasizes relations among actors and the quality of the dialogue." Through the interaction of the various stakeholders involved in the transition process at the municipal and metropolitan levels, some interesting initiatives have evolved and social learning processes have taken place at the policy level.

Following the framework of Reed et al. (2010) about the three main requirements for social learning processes to take place successfully, we argue that while much learning has been generated during the transition process, especially at technical and conceptual levels, social learning processes are still lagging behind. Important changes and learning have been generated due to the interaction between the different stakeholders involved in the policy process which have helped to modify the framework of ideas and basic assumptions that some actors support. In this way, social learning processes have materialized at the policy level to some extent. However, users and the rest of society have for the most part been excluded from this process. Users mostly learn about greywater reuse through experimentation by having the greywater reuse system installed in their apartments. The second prerequisite of social learning has not been completely fulfilled yet as knowledge about greywater reuse benefits and requirements have rarely penetrated through the whole community. Users usually learn about

the new technology in small groups. Several apartments (at least 8) usually share the same greywater reuse system but this is only limited to new apartment buildings which at the moment are still small in number in most communities. The statement of one of the environmental managers interviewed stresses the importance of stimulating social learning processes among citizens for the success of greywater reuse technologies: "we supervise everything that is visible but the rest is a matter of faith" (Interview #2, 30 August 2010). As Ison and Watson (2007) point out creating purposefully the conditions for social learning is still a major challenge.

Different local, metropolitan and regional governmental actors have to some extent acted as "transition managers" using the terminology of Loorbach and Rotmans (2006). The transition manager is responsible for bringing together the different actors involved in the transitions arena (van der Kerkhof and Wieczorek, 2005). However, as Rotmans et al. (2001, p. 25) point out governments cannot act as "the great commander, enforcing change". This statement is in line with the reflection of one of the environmental managers interviewed: "I intuit that many systems are not properly maintained and have been by-passed and we should penalize users for this reason but a fine would end with the water saving ordinance" (Interview #3, 1 September 2010). This reflection is a good reminder of the critical importance of social change for transition management and regime shifts.

6. Conclusions

Greywater reuse and its presence in the local water ordinances passed in several municipalities of the Barcelona area has been used in this paper as a lens to analyze the possibilities and barriers toward the generalization of a new socio-technical urban water regime based on decentralized water technologies. In the first steps of this transition, especially at the niche level, much technical learning has been generated by trial and error and more sophisticated and efficient technologies have displaced less sophisticated ways of dealing with greywater. Policies have also suffered continuous modifications in order to adapt to the new realities and incorporate the conceptual learning generated. Interactions and communication between actors have also proved to be essential to generate social learning processes among policy-makers.

We have presented the case of Sant Cugat del Vallès, arguing that technological advances at the niche level, especially the use of membranes, have eased the early concerns about quality and health issues of greywater systems based on chemical products (chlorine). A critical aspect in the social acceptance process of these new technologies appears to be the reliability and know-how of professional maintenance companies which were somewhat problematic in the early days of the "chlorine pill" but that have gained more consistence with the increasing use of membranes. Thus, technological breakthroughs, such as the increasing use of membranes may reinforce a socio-technical regime based on decentralized water systems. However, not only experimentation and technical improvements are needed for the transition to take-off, stakeholder coordination and mindset changes are equally important and even more difficult to achieve.

While greywater may gain acceptance at the local level with the new membrane technologies on the market and therefore increase its participation in the transition toward more sustainable water management, the promotion of this socio-technical system at the regional and national levels appears more problematic. The various research studies funded by different regional institutions have been instrumental in moving the debate forward and enhancing the collaboration between the different stakeholders involved. However, greater efforts in this direction are needed to truly regionalize and secure the transition. Some environmental managers proposed the creation of a regional entity as a promising solution to ensure the safe operation of greywater systems and more affordable and viable water quality testing. Setting homogeneous rules and standards regarding for instance water quality is also necessary. Unless these standards are set and approved by the competent authorities, new investments and new technological developments could be hard to implement. Social learning processes among users and more importantly, among regular citizens also need to take place but this change is more difficult to promote. Future advances in the technical and policy front will certainly benefit social learning processes at the grass-root level.

Rotmans et al. (2001, p. 25) stress that governments should not act as "the great commander, enforcing change." This case study has attempted to show that local regulations enforced by local

governments may be an important element to accelerate the pace of sustainability transitions. However, we also recognize that regulations alone cannot guarantee a successful transition. If the learning required is not generated in a timely manner and the actors involved do not cooperate, the transition can easily collapse. Changes have to take place at different levels to allow a socio-technical sustainability transition to take hold. For example, the whole process would probably benefit from a more direct involvement of supralocal government bodies (at the regional and national levels).

It is also important to remember that niches and regimes are embedded in a landscape. External factors such as economic crisis, droughts and scarcity episodes, or political changes (at local, Catalan, Spanish or European levels) can challenge the stability of a given socio-technical regime (e.g. the centralized water model of "flush and forget") and enable the transition toward a more sustainable model; or the other way round. A recent hurdle for scaling-up decentralized systems has been the political change in the regional government after the elections of late 2010. Given the critical financial condition of the Catalan Water Agency, the new regional government has dismantled many of the initiatives regarding new water resources undertaken by the previous administration. Greywater has thus ceased to be a priority for the regional water authority and the scaling-up of this resource in any of the fields where much and relevant work had been done (quality issues, for instance) now appears very problematic. On the other hand, in 2011 the Spanish Conservative Party, the same party that approved the National Water Plan of 2001, came back to power. While a turn toward the hydraulic paradigm, involving the increase of the supply, via large infrastructures such as transfers, has been already hinted in some river basin plans, it remains to be seen whether deeper changes in Spanish water politics will occur in the future.

In conclusion, urban water supply may increasingly become characterized by a combination of different sources. In certain areas such as Barcelona, traditional large scale supply systems (reservoirs, inter-basin transfers), which have been articulating the hegemonic water regime in the last century, appear to have reached important economic, social and environmental limits. The recent drought episodes combined with the emergence of a New Water Culture at the Spanish level have supposed the mutation of this regime. A growing number of municipalities are using more local and decentralized resources such as rainwater or greywater, which are challenging the essence of the hydraulic paradigm. Greywater implementation in metropolitan Barcelona shows the uneven speeds of the transition to more sustainable urban water regimes, as well as the hurdles and the spaces of opportunity and hope it may encounter.

The urban water sector seems to have ideal characteristics to be a model system for transitions studies, as it combines ecological, economic, technological and social aspects that may appeal the transitions research community. We would like to end however with a critical self-reflection on greywater reuse and its ambiguous character toward the incumbent hydraulic regime. While it radically deviates from the regime in some aspects, such as the non-reliance on direct water abstraction from rivers or aquifers, it may also reproduce it unintentionally in other aspects. It might do so in being a technical solution which requires only minimal adjustment from the consumer and an apparent strong focus on water quality standards. This does not mean that this technology is not valid for transition studies but rather to be aware that most radical innovations have ambiguous relations with the existing water regime. Finally, we must be aware of the relatively limited scope of these alternative resources. While the potential is indeed high, greywater reuse remains in quantitative terms a rather marginal resource in the whole water supply architecture of the Barcelona area. Still, in qualitative terms, it represents an important component of new socio-technical realities at the household level regarding water use and management.

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